

genaut — frontier report

source CSV : genaut/logs/2statelap0acc/default.csv
corpus rows: 30

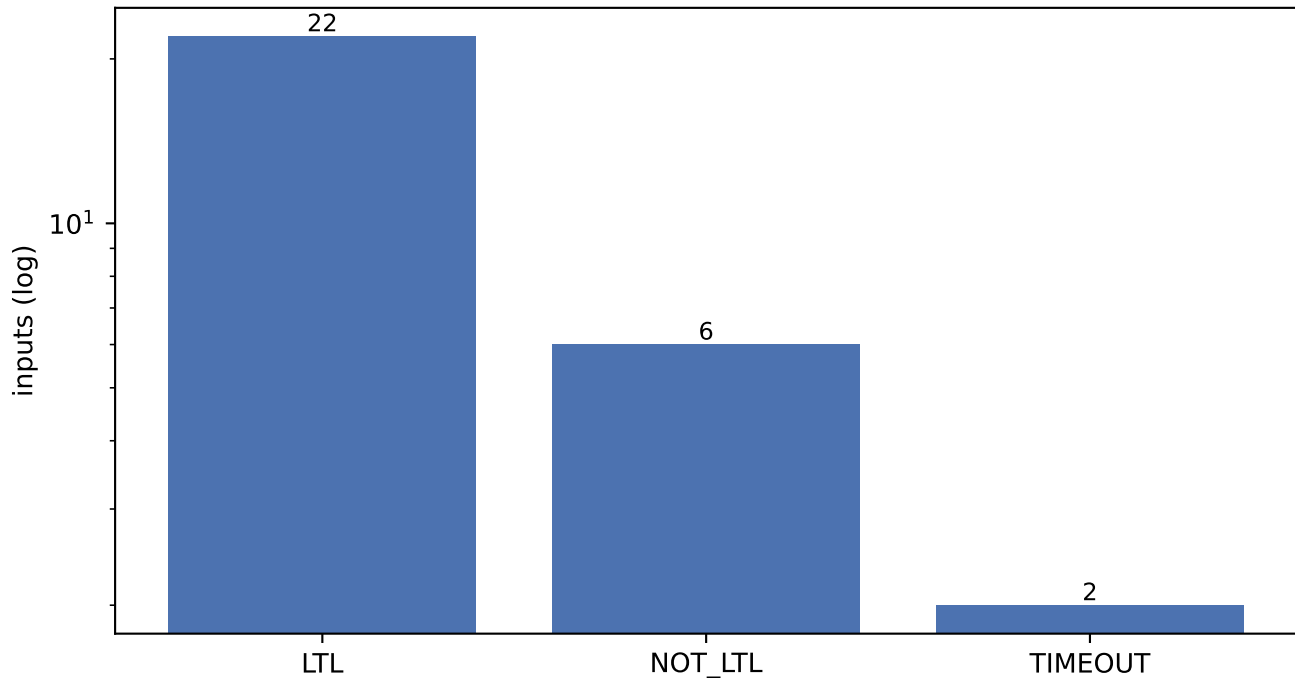
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – structural (md5) concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 30
  LTL-definable (formula built) :    22  ( 73.3%)
    not-LTL (decided)           :     6  ( 20.0%)
    ambiguous (timeout, unknown) :     2  (  6.7%)
decided (LTL or not-LTL): 28  ( 93.3%)
```

Answer frontier (by result)



Formula collapse — structural (md5) concentration

LTL answers: 22 (0 with a gated/unprintable string)

distinct structures (md5): 17

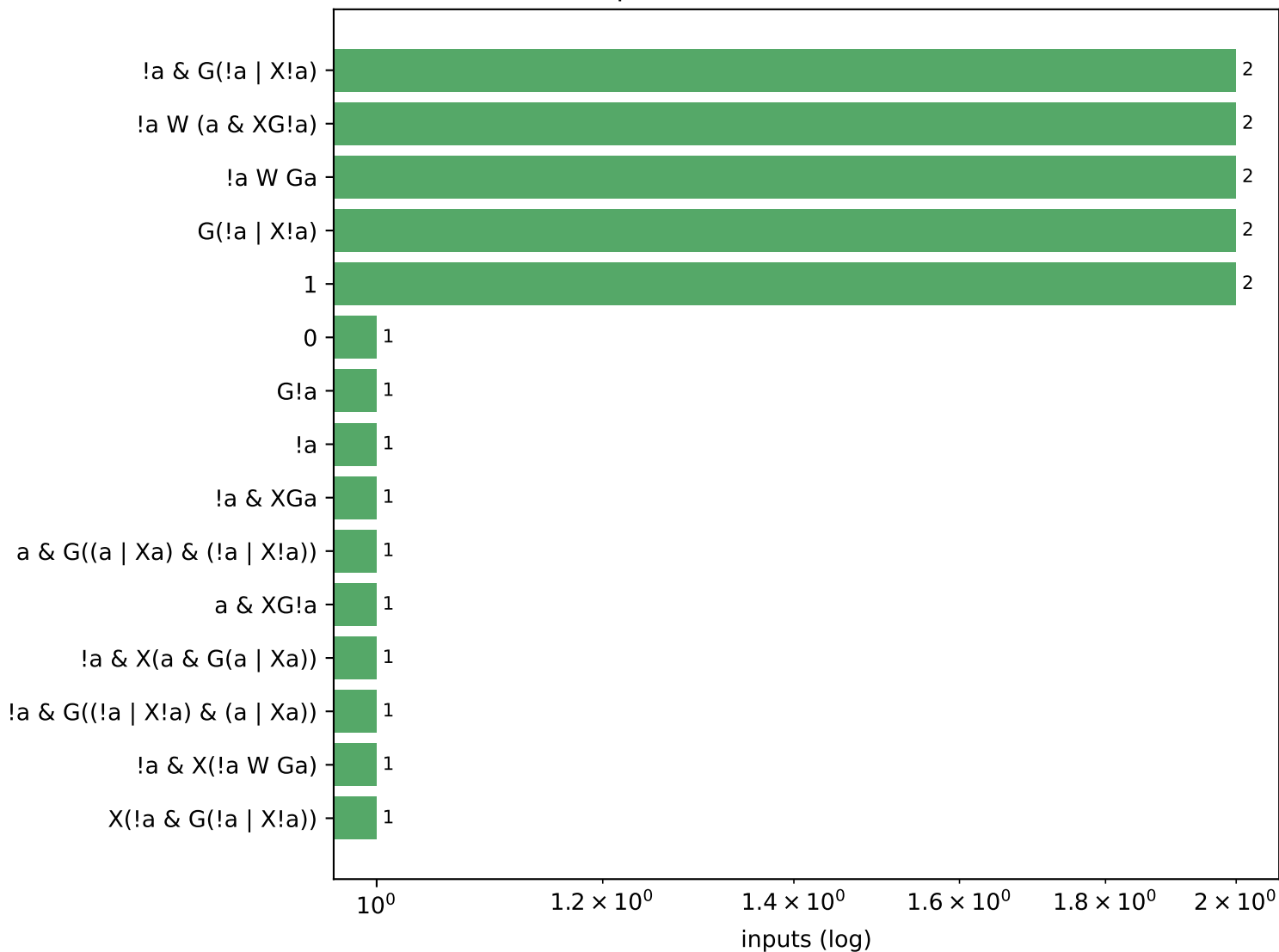
80% of answers covered by the top 13 structure(s)

90% of answers covered by the top 15 structure(s)

top 15 structures:

```
2 ( 9.1%) !a & G(!a | X!a)
2 ( 9.1%) !a W (a & XG!a)
2 ( 9.1%) !a W Ga
2 ( 9.1%) G(!a | X!a)
2 ( 9.1%) 1
1 ( 4.5%) 0
1 ( 4.5%) G!a
1 ( 4.5%) !a
1 ( 4.5%) !a & XGa
1 ( 4.5%) a & G((a | Xa) & (!a | X!a))
1 ( 4.5%) a & XG!a
1 ( 4.5%) !a & X(a & G(a | Xa))
1 ( 4.5%) !a & G(!a | X!a) & (a | Xa)
1 ( 4.5%) !a & X(!a W Ga)
1 ( 4.5%) X(!a & G(!a | X!a))
```

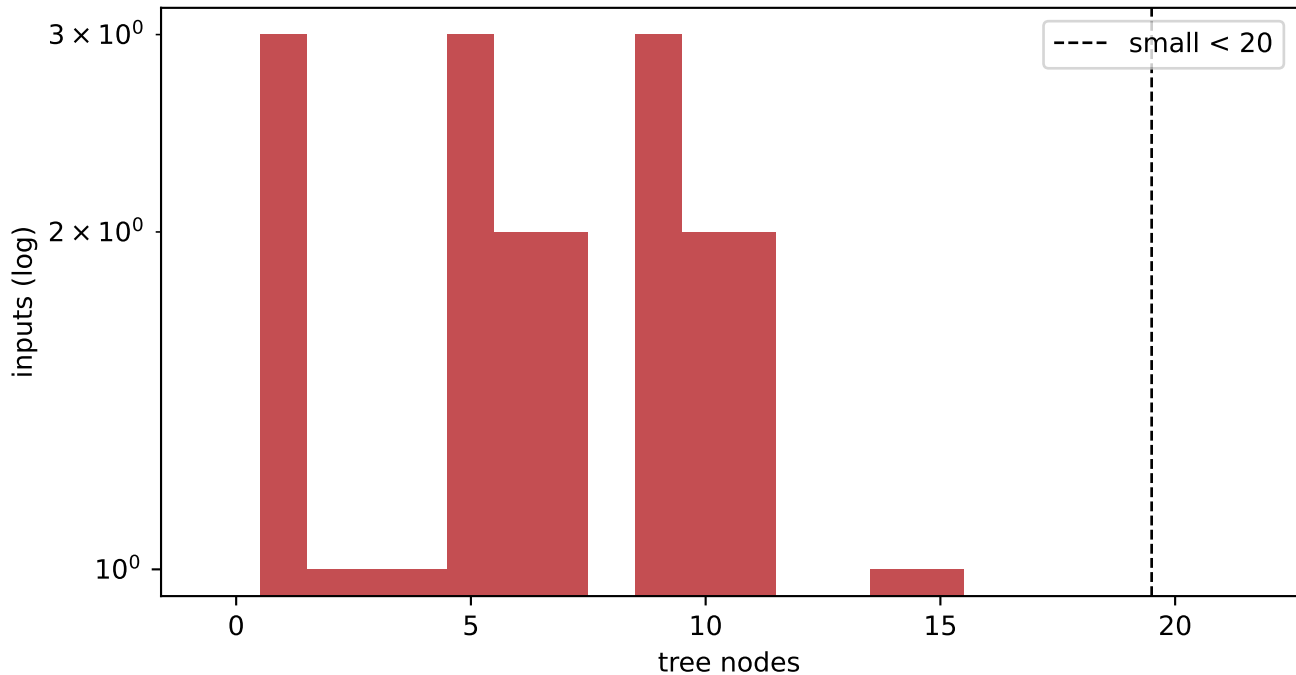
Top 15 reconstructed structures



Size frontier — flattened formula size

```
LTL answers with a size: 22
trivial (tree<=1, i.e. true/false): 3 ( 13.6%)
small   (tree<20):                  22 (100.0%)
larger  (tree>=20):                  0 (  0.0%)
tree nodes: median 6  mean 6.9  max 15
```

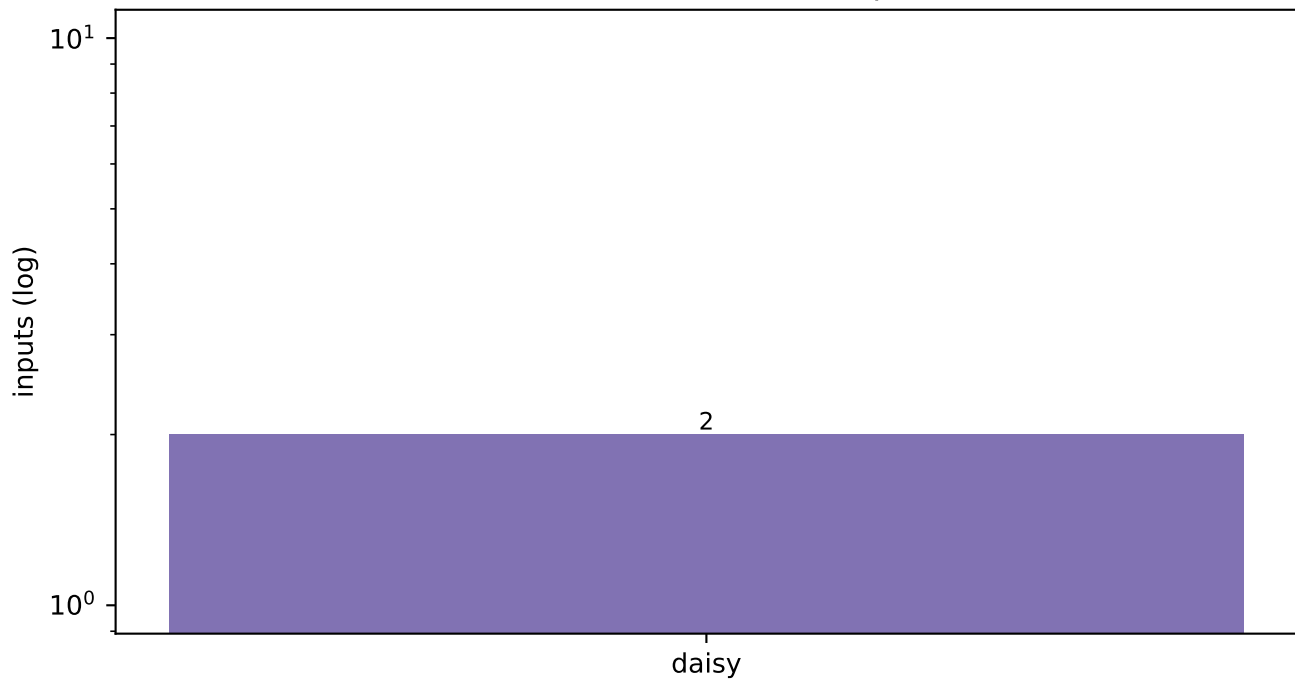
Flattened size (tree nodes); clipped at 21 (0 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 2  
  2 (100.0%) daisy  
build_s over true rows: total 0.1s  mean 0.071s  max 0.073s
```

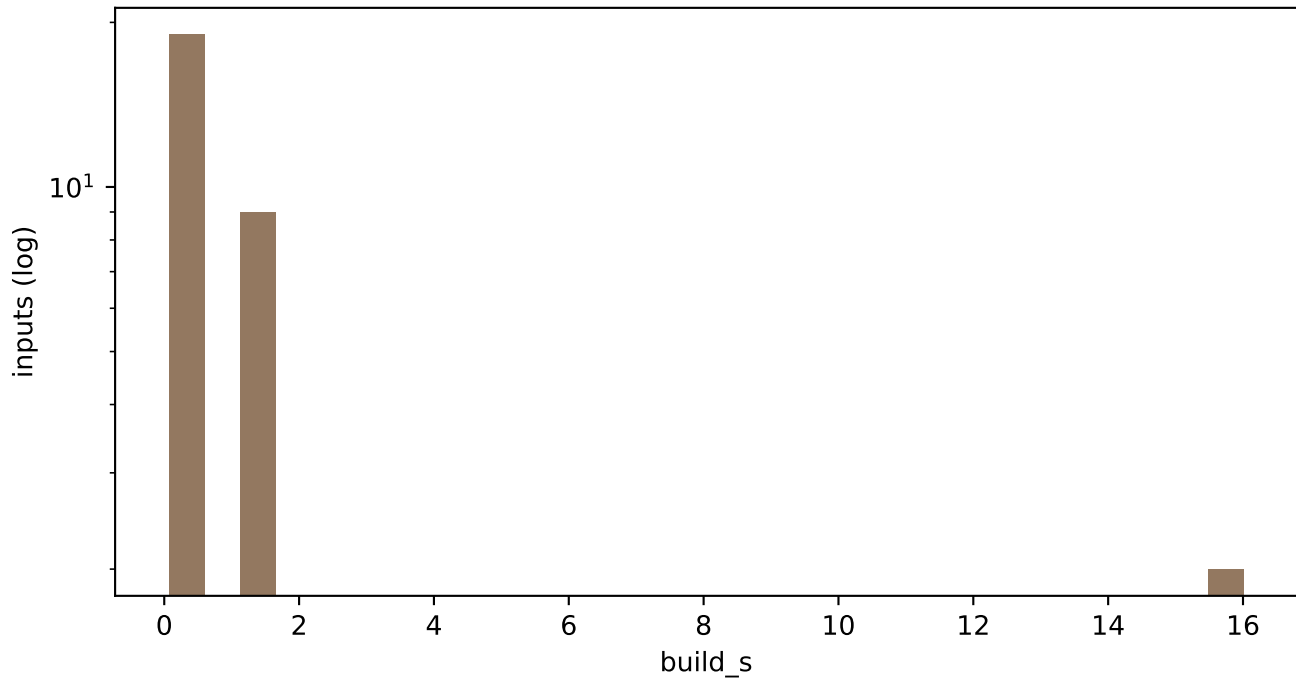

Routes to `true` (technique)



Build time & timeouts

```
total build: 44.3s   mean 1.478s   max 16.007s
  LTL           n= 22   mean 0.234s   max 1.273s
  NOT_LTL       n=  6   mean 1.198s   max 1.216s
timeouts: 2
  2statelap0acc_086.hoa
  2statelap0acc_106.hoa
```

Build-time distribution



Verification — spot-oracle validation

validation over 22 LTL answers:
True 22 (100.0%)
NOT-equivalent (FALSE): 0 CLEAN